THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 77	Sum $= 11(a+b)$.
2	**(a)** 63	Difference $= 9(a-b)$.
3	**(a)** 5	List the terms and count the steps.
4	**(a)** 2	Digit sum must be a multiple of 9.
5	**(d)** 55	Triangular number $\frac{n(n+1)}{2}$.
6	**(c)** 0 or 5	Recall the divisibility test.
7	**(b)** 40	Pairs around the middle cancel out.
8	**(b)** 7531	Write the digits in decreasing order.
9	**(a)** 99	Sum $= 11(a+b)$.
10	**(c)** 36	Difference $= 9(a-b)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$10a+b+10b+a = 11a+11b; = 11(a+b)$ (1); For $a=3, b=5$: $35+53=88=11\times 8$ (1).	1+1
12	Known digit sum $= 13$; Total $= (13+*)$ must be a multiple of 9 (1); $* = 5$; Check: 4527 has digit sum 18 (1).	1+1
13	Middle number $= 110 \div 5 = 22$; The numbers are $18, 20, 22, 24, 26$ (1); Check: sum $= 110$ (1).	1+1
14	$10a+b+10b+a = 11a+11b; = 11(a+b)$ (1); For $a=4, b=7$: $47+74=121=11\times 11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: $9652-2569=7083$ (1); Step 2: $8730-0378=8352$ (1); After two steps the number is 8352 (1).	1+1+1
16	$57+75=132$; $132+231=363$ (1); Number of steps $= 2$ (1); Palindrome: 363 (1).	1+1+1
17	Sum of three consecutive numbers $= 3\times$ middle (1); $63 \div 3 = 21$ is a whole number, so yes (1); The numbers are $20, 21, 22$ (1).	1+1+1
	OR: Step 1: $7521-1257=6264$ (1); Step 2: $6642-2466=4176$ (1); After two steps the number is 4176 (1).	1+1+1
18	$49+94=143$; $143+341=484$ (1); Number of steps $= 2$ (1); Palindrome: 484 (1).	1+1+1
	OR: Sum of three consecutive numbers $= 3\times$ middle, a multiple of 3 (1); Digit sum of 100 is 1, not a multiple of 3 (1); 100 is not a multiple of 3, so it cannot be written in this way (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 237$ (1); Difference $= 732-237=495$ (1); $= 99\times 5$ (1); In general $100a+c-100c-a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 9$ (1); Triple-and-add-1 steps $= 3$ (1); Total steps $= 12$ (2).	1+1+1+2
20	Reverse $= 549$ (1); Difference $= 945-549=396$ (1); $= 99\times 4$ (1); In general $100a+c-100c-a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 14$ (1); Triple-and-add-1 steps $= 6$ (1); Total steps $= 20$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a+b$ as $10a+b$ again; the reverse is $10b+a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 8	List the terms and count the steps.
2	**(d)** 6	Digit sum must be a multiple of 9.
3	**(b)** 210	Triangular number $\frac{n(n+1)}{2}$.
4	**(c)** 0, 2, 4, 6 or 8	Recall the divisibility test.
5	**(d)** 65	Pairs around the middle cancel out.
6	**(a)** 8420	Write the digits in decreasing order.
7	**(d)** 132	Sum $= 11(a+b)$.
8	**(d)** 45	Difference $= 9(a-b)$.
9	**(a)** 16	List the terms and count the steps.
10	**(a)** 5	Digit sum must be a multiple of 9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Known digit sum $= 15$; Total $= (15+*)$ must be a multiple of 9 (1); $* = 3$; Check: 7335 has digit sum 18 (1).	1+1
12	Middle number $= 180 \div 5 = 36$; The numbers are $32, 34, 36, 38, 40$ (1); Check: sum $= 180$ (1).	1+1
13	$10a+b+10b+a = 11a+11b$; $= 11(a+b)$ (1); For $a=2, b=9$: $29+92 = 121 = 11\times 11$ (1).	1+1
14	Known digit sum $= 18$; Total $= (18+*)$ must be a multiple of 9 (1); $* = 8$; Check: 18962 has digit sum 26 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: $8420-0248 = 8172$ (1); Step 2: $8721-1278 = 7443$ (1); After two steps the number is 7443 (1).	1+1+1
16	$59+95 = 154$; $154+451 = 605$; $605+506 = 1111$ (1); Number of steps $= 3$ (1); Palindrome: 1111 (1).	1+1+1
17	Sum of three consecutive numbers $= 3\times$ middle (1); $111 \div 3 = 37$ is a whole number, so yes (1); The numbers are $36, 37, 38$ (1).	1+1+1
	OR: Step 1: $8620-0268 = 8352$ (1); Step 2: $8532-2358 = 6174$ (1); After two steps the number is 6174 (1).	1+1+1
18	$38+83 = 121$ (1); Number of steps $= 1$ (1); Palindrome: 121 (1).	1+1+1
	OR: Sum of three consecutive numbers $= 3\times$ middle, a multiple of 3 (1); Digit sum of 145 is 10, not a multiple of 3 (1); 145 is not a multiple of 3, so it cannot be written in this way (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 318$ (1); Difference $= 813-318 = 495$ (1); $= 99\times 5$ (1); In general $100a+c-100c-a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $21, 64, 32, 16, 8, 4, 2, 1$ (1); Halvings $= 6$ (1); Triple-and-add-1 steps $= 1$ (1); Total steps $= 7$ (2).	1+1+1+2
20	Reverse $= 156$ (1); Difference $= 651-156 = 495$ (1); $= 99\times 5$ (1); In general $100a+c-100c-a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 11$ (1); Triple-and-add-1 steps $= 4$ (1); Total steps $= 15$ (2).	1+1+1+2

Mistakes to avoid:

- Saying the sum of two odd numbers is odd; it is even.
- Treating the product of two odd numbers as even; it is odd.
- Writing the reverse of $10a+b$ as $10a+b$ again; the reverse is $10b+a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
- In Kaprekar's routine, forgetting to keep leading zeros when the smaller arrangement starts with 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 465	Triangular number $\frac{n(n+1)}{2}$.
2	**(d)** 0	Recall the divisibility test.
3	**(d)** 100	Pairs around the middle cancel out.
4	**(b)** 9631	Write the digits in decreasing order.
5	**(d)** 143	Sum $= 11(a+b)$.
6	**(a)** 27	Difference $= 9(a-b)$.
7	**(b)** 6	List the terms and count the steps.
8	**(b)** 5	Digit sum must be a multiple of 9.
9	**(c)** 820	Triangular number $\frac{n(n+1)}{2}$.
10	**(b)** its digit sum is divisible by 3	Recall the divisibility test.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Middle number $= 70 \div 5 = 14$; The numbers are $10, 12, 14, 16, 18$ (1); Check: sum $= 70$ (1).	1+1
12	$10a + b + 10b + a = 11a + 11b$; $= 11(a+b)$ (1); For $a = 6, b = 1$: $61 + 16 = 77 = 11 \times 7$ (1).	1+1
13	Known digit sum $= 14$; Total $= (14 + *)$ must be a multiple of 9 (1); $* = 4$; Check: 5841 has digit sum 18 (1).	1+1
14	Middle number $= 250 \div 5 = 50$; The numbers are $46, 48, 50, 52, 54$ (1); Check: sum $= 250$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Step 1: $9531 - 1359 = 8172$ (1); Step 2: $8721 - 1278 = 7443$ (1); After two steps the number is 7443 (1).	1+1+1
16	$18 + 81 = 99$ (1); Number of steps $= 1$ (1); Palindrome: 99 (1).	1+1+1
17	Sum of three consecutive numbers $= 3\times$ middle (1); $150 \div 3 = 50$ is a whole number, so yes (1); The numbers are $49, 50, 51$ (1).	1+1+1
	OR: Step 1: $7631 - 1367 = 6264$ (1); Step 2: $6642 - 2466 = 4176$ (1); After two steps the number is 4176 (1).	1+1+1
18	$26 + 62 = 88$ (1); Number of steps $= 1$ (1); Palindrome: 88 (1).	1+1+1
	OR: Sum of three consecutive numbers $= 3\times$ middle, a multiple of 3 (1); Digit sum of 98 is 17, not a multiple of 3 (1); 98 is not a multiple of 3, so it cannot be written in this way (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reverse $= 409$ (1); Difference $= 904 - 409 = 495$ (1); $= 99 \times 5$ (1); In general $100a + c - 100c - a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $24, 12, 6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 8$ (1); Triple-and-add-1 steps $= 2$ (1); Total steps $= 10$ (2).	1+1+1+2
20	Reverse $= 125$ (1); Difference $= 521 - 125 = 396$ (1); $= 99 \times 4$ (1); In general $100a + c - 100c - a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: Sequence: $25, 76, 38, 19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 16$ (1); Triple-and-add-1 steps $= 7$ (1); Total steps $= 23$ (2).	1+1+1+2

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- Writing the reverse of $10a + b$ as $10a + b$ again; the reverse is $10b + a$.
- Using the divisibility test for 9 as if it were for 3 (digit sum 12 is divisible by 3 but not 9).
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